

Toward a Next-Generation First-Principles Theory of Electron–Phonon Coupling

Why DFPT Works, Where It Fails, and a Channel-Resolved Correction Strategy

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Abstract

Density-functional perturbation theory (DFPT) is the standard first-principles method for lattice dynamics and electron–phonon coupling. Its practical success extends from simple metals and semiconductors to several materials with appreciable electronic correlations, even though the physical scattering amplitude is a many-body vertex and DFPT uses a static Kohn–Sham response. The central question of this proposal is therefore not whether corrections to DFPT exist, but why they are often small for some phonons and large for others.

We formulate a testable working hypothesis: the DFPT error is controlled primarily by the operator channel in which a phonon perturbs the low-energy electronic Hamiltonian, together with the channel-resolved many-body susceptibility and its frequency dependence. A common charge-like perturbation is constrained in the strict uniform limit and has a particularly simple realization in the uniform electron gas. Orbital splitting, hybridization, hopping, spin, and interaction modulation are not protected by the same constraint. This distinction explains why strong correlation alone is not a sufficient failure criterion and motivates a next-generation method: preserve DFPT as a low-cost baseline, project each phonon perturbation into physically defined channels, and add only the static or dynamic many-body corrections required by those channels. The proposal specifies the equations, limits, benchmark materials, falsification criteria, a two-day literature sprint, and an AI-agent workflow for turning the hypothesis into a transparent prediction framework.

Proposed deliverable. The intended result is not a universal scalar multiplier for every electron–phonon matrix element. It is a channel-resolved, uncertainty-aware correction layer on top of DFPT that can also decide when uncorrected DFPT is already adequate.

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1 Problem statement and scope

In crystalline solids, the interaction between electrons and lattice vibrations controls electrical resistance, conventional superconductivity, thermoelectric transport, carrier relaxation, and device heating. DFPT is the most widely used first-principles tool for calculating phonons and electron–phonon matrix elements because it avoids a separate large-supercell calculation for every displacement and treats the self-consistent electronic response directly [Baroni et al., 2001, Gonze, 1995, Giustino, 2017].

The motivating observation of Quantum Harness Issue 35 is that DFPT is often quantitatively useful in a surprisingly broad range of materials [QuantumBFS, 2026]. This statement needs careful wording. It does *not* mean that ordinary semilocal DFPT is uniformly accurate for every correlated material, every phonon, and every observable. Standard DFPT can fail when the reference electronic structure is qualitatively wrong, when a phonon couples to an unprotected internal degree of freedom, or when the electronic response is strongly frequency dependent. The real puzzle is narrower and more useful:

Core scientific question. Why are corrections beyond static Kohn–Sham linear response small for some electron–phonon perturbations, including some perturbations in correlated materials, while they are large for other modes in the same material?

This formulation rules out two incomplete answers.

1. **“The quasiparticle residue cancels the vertex.”** A numerical product close to one is evidence, not a physical explanation. The Ward identity produces such a cancellation only in a specific conserved-charge limit, whose order of limits is not the generic adiabatic phonon limit.
2. **“DFPT works because the phonon shifts every electronic state by the same amount.”** Real perturbations do not act equally on all Bloch states. Even a local operator proportional to the identity inside a correlated orbital subspace produces band- and momentum-dependent matrix elements because Bloch states have different orbital weights and phases.

The proposal instead treats the derivative of the electronic Hamiltonian as an operator. It asks which part of that operator couples to total charge, which part splits or mixes internal orbitals, and which part changes hopping, hybridization, or interaction parameters. Conservation laws constrain the first category in a special limit; they do not constrain all categories equally.

1.1 Target deliverable

The target is a computational framework that uses DFPT as a baseline and adds a small number of physically interpretable, channel-resolved many-body corrections. It should be:

- **simpler** than a full many-body calculation of every $g_{mn\nu}(\mathbf{k}, \mathbf{q})$;
- **faster** by focusing expensive calculations on risky modes and low-dimensional operator channels;
- **transparent** because every correction is attached to a named physical operator and a documented validity test;

- **falsifiable** on materials for which DFPT succeeds for one mode but fails for another; and
- **uncertainty-aware**, including an abstention path when a mode lies outside the validated regime.

The first validation object may be the uniform electron gas (UEG), but it is not the final theory. The UEG contains only a scalar density channel and therefore provides a clean baseline for understanding what is special about that channel. The proposed method must then survive multi-orbital and correlated-material tests.

1.2 What is and is not being corrected

DFPT computes several related quantities: force constants, phonon eigenvectors, Born effective charges, dielectric response, and electron–phonon matrix elements. The main target here is the last quantity: the physical quasiparticle scattering vertex associated with a phonon. Corrections to equilibrium structure, anharmonicity, or the phonon spectrum may also matter, but they are separate axes of error. The proposed benchmark will keep the following gates distinct:

1. the Kohn–Sham ground-state and phonon baseline;
2. the electron–phonon matrix element g ;
3. the observable assembled from g , such as a linewidth, mobility, or pairing kernel.

Passing one gate does not establish the next.

2 What standard DFPT actually assumes

2.1 The perturbation and the matrix element

Let $u_{\nu\mathbf{q}}$ be the normal coordinate of phonon branch ν and wavevector $\mathbf{q}$. After including the conventional phonon normalization factor, define the self-consistent Kohn–Sham perturbation operator

$$D_{\nu\mathbf{q}}^{\text{KS}} \equiv \partial_{\nu\mathbf{q}} V_{\text{KS}} = \partial_{\nu\mathbf{q}} V_{\text{ion}} + \partial_{\nu\mathbf{q}} V_{\text{Hxc}}. \quad (1)$$

The DFPT electron–phonon matrix element is

$$g_{mn\nu}^{\text{DFPT}}(\mathbf{k}, \mathbf{q}) = \langle \psi_{m,\mathbf{k}+\mathbf{q}} | D_{\nu\mathbf{q}}^{\text{KS}} | \psi_{n,\mathbf{k}} \rangle. \quad (2)$$

The perturbation is a matrix in band, orbital, spin, and reciprocal-lattice indices. Therefore standard DFPT does not assume that every electron sees the same scalar shift.

The first-order orbital response is obtained from a Sternheimer equation of the schematic form

$$(\hat{H}_{\text{KS}} - \varepsilon_{n\mathbf{k}}) |\partial_{\nu\mathbf{q}} \psi_{n\mathbf{k}}\rangle = -\hat{Q}_{\mathbf{k}+\mathbf{q}} (\partial_{\nu\mathbf{q}} V_{\text{KS}} - \partial_{\nu\mathbf{q}} \varepsilon_{n\mathbf{k}}) |\psi_{n\mathbf{k}}\rangle, \quad (3)$$

where $\hat{Q}$ projects outside the occupied state or selected variational subspace. The induced density updates V_{Hxc} until self-consistency.

2.2 Static electronic equilibrium

In ordinary adiabatic DFPT, the electrons are assumed to remain in the instantaneous ground state associated with a slowly displaced lattice [Gonze, 1995, Baroni et al., 2001]. Equivalently, the

electronic response entering [equation \(3\)](#) is evaluated at zero external frequency. The electronic denominator contains Kohn–Sham energy differences, but not a phonon-frequency shift such as $\varepsilon_{m,\mathbf{k}+\mathbf{q}} - \varepsilon_{n\mathbf{k}} \pm \omega_{\nu\mathbf{q}}$.

This assumption is usually sensible when electronic relaxation and interband energy scales are fast compared with the phonon period. It can fail for low carrier densities, narrow bands, small Fermi energies, polar optical modes, collective modes, or any situation in which $\omega_{\nu\mathbf{q}}$ is not negligible on the scale governing the electronic response.

2.3 The response equation behind DFPT screening

For a scalar density perturbation, suppressing matrix indices for clarity,

$$\delta V_{\text{KS}} = \delta V_{\text{ion}} + (v + f_{\text{xc}})\delta n, \quad (4)$$

$$\delta n = \chi_s \delta V_{\text{KS}}, \quad (5)$$

where v is the Coulomb interaction, f_{xc} is the static exchange–correlation kernel, and χ_s is the independent Kohn–Sham density response. Solving the two equations gives

$$\delta V_{\text{KS}} = [1 - (v + f_{\text{xc}})\chi_s]^{-1} \delta V_{\text{ion}}, \quad g^{\text{DFPT}} = \frac{g^0}{1 - (v + f_{\text{xc}})\chi_s} \quad (6)$$

in a translationally invariant scalar problem. In a crystal these quantities are matrices in reciprocal lattice vectors and may also carry spin or sublattice indices.

2.4 The precise missing term

Choose the convention

$$G^{-1}(\omega) = \omega + \mu - H_{\text{KS}} - \Sigma^{\text{corr}}(\omega), \quad (7)$$

where Σ^{corr} is the self-energy correction beyond the chosen Kohn–Sham reference, with the corresponding double counting removed. The many-body perturbation of the quasiparticle equation contains

$$D_{\nu\mathbf{q}}^{\text{MB}}(\omega) = \partial_{\nu\mathbf{q}} [V_{\text{KS}} + \Sigma^{\text{corr}}(\omega)]. \quad (8)$$

Thus the central vertex term absent from standard DFPT is $\partial_{\nu\mathbf{q}}\Sigma^{\text{corr}}$, together with its frequency dependence and consistent wavefunction renormalization. A large equilibrium self-energy does not logically imply a large derivative with respect to a particular phonon. This distinction is the first reason why the label “strongly correlated” alone cannot predict the DFPT error. A consistent separation between the DFT starting point and many-body electron–phonon corrections is discussed in detail by [Marini et al. \[2015\]](#).

Operational limitation of DFPT. Standard DFPT is a static Kohn–Sham response. Its reliability for a physical quasiparticle vertex depends on whether the phonon derivative of the beyond-DFT self-energy is small, already represented by the chosen static density functional, or cancelled by a constraint in the relevant operator channel.

3 Many-body response, screening, and conservation laws

3.1 Why screening, polarization, and the vertex are not independent

In Hedin's closed set of equations, the irreducible polarization P , dielectric function ϵ , screened interaction W , and reducible density response χ satisfy schematically [Hedin, 1965, Giustino, 2017]

$$P = -iGG\Gamma_\rho, \quad (9)$$

$$\epsilon = 1 - vP, \quad (10)$$

$$W = \epsilon^{-1}v, \quad (11)$$

$$\chi = P(1 - vP)^{-1}. \quad (12)$$

Here $\Gamma_\rho = -\delta G^{-1}/\delta V_{\text{tot}}$ is a proper density vertex in the stated convention. Matrix products and integrations over internal variables are implicit. These relations answer an important conceptual question: screening, the quasiparticle vertex, and static density response are not three independent physical factors available for arbitrary multiplication. The vertex enters P ; P determines ϵ ; and χ follows from both. Mixing proper and screened vertex conventions can therefore count screening twice.

In a scalar electron gas, a convenient schematic form for the quasiparticle coupling to an external ionic perturbation is

$$g^{\text{phys}}(\mathbf{q}, \omega) = g^0(\mathbf{q}) \frac{z\Gamma_\rho(\mathbf{q}, \omega)}{\epsilon(\mathbf{q}, \omega)}, \quad (13)$$

where z is the quasiparticle residue. This equation is useful only after the vertex convention and screening factor have been stated.

3.2 The Ward identity and its two limits

Charge conservation and gauge invariance imply the Ward–Takahashi identity [Nambu, 1960]

$$i\nu\Gamma^0(k+q, k) - \mathbf{q} \cdot \mathbf{\Gamma}^J(k+q, k) = G^{-1}(k+q) - G^{-1}(k), \quad (14)$$

up to sign conventions for the Fourier transform and electric charge. Γ^0 is the scalar charge vertex and $\mathbf{\Gamma}^J$ is the current vertex. The identity constrains the full four-vector vertex; the conclusion depends on how $(\mathbf{q}, \nu)$ approaches zero.

If one sets $\mathbf{q} = 0$ first and then takes $\nu \rightarrow 0$, the dynamic uniform limit is

$$\Gamma_\omega^0 = \frac{\partial G^{-1}}{\partial(i\omega)} = 1 - \frac{\partial \Sigma}{\partial(i\omega)} = z^{-1}, \quad z\Gamma_\omega^0 = 1. \quad (15)$$

This is an exact cancellation for the conserved total charge in that limit.

If instead one sets $\nu = 0$ first and then lets $\mathbf{q} \rightarrow 0$, the Ward identity directly constrains the longitudinal current vertex. The static scalar response also contains the response of the self-energy to chemical potential and the Landau interaction. For an isotropic Fermi liquid, for example,

$$\frac{\partial n}{\partial \mu} = \frac{N^*(0)}{1 + F_0^s}, \quad (16)$$

not simply $zN_0(0)$. Therefore $z\Gamma_\rho = 1$ is not a universal static density-vertex identity.

For an acoustic phonon, $\omega_{\nu\mathbf{q}} \simeq c_s|\mathbf{q}|$ and generally $v_F/c_s \gg 1$, so

$$\frac{v_F|\mathbf{q}|}{\omega_{\nu\mathbf{q}}} \simeq \frac{v_F}{c_s} \gg 1. \quad (17)$$

This is the adiabatic or static side of the response. The dynamic result in equation (15) cannot by itself explain the success of ordinary DFPT for acoustic phonons.

3.3 The exact common-potential anchor

There is nevertheless a clear physical picture in the strict uniform case. If a lattice displacement produces a potential $H' = Du\hat{N}$ that couples only to the conserved total particle number, every many-electron state with N electrons shifts by NDu . The energy required to add one electron therefore shifts by exactly Du , irrespective of the strength of interactions inside the system.

Equivalently, a common scalar shift can be absorbed into the frequency origin,

$$\Sigma_u(\omega) = \Sigma_0(\omega - Du), \quad \partial_u \Sigma = -D \partial_\omega \Sigma. \quad (18)$$

The shift of the quasiparticle pole is then

$$z[D + \partial_u \Sigma] = D. \quad (19)$$

This derivation explains the cancellation physically: an interaction cannot renormalize a change of the energy zero. It is an exact anchor at strict $\mathbf{q} = 0$ for a common charge potential, not a proof at finite momentum and not a statement about orbital splitting, bond modulation, or other internal operators.

What conservation does for this project. Conservation laws identify a protected direction in the space of perturbation operators. The research task is to measure how much of each real phonon lies along that direction and how strongly the remaining, unprotected directions are renormalized.

4 What the uniform electron gas does and does not explain

4.1 Uniform equilibrium does not mean only zero momentum

The UEG has a constant equilibrium density, but it can be perturbed at every wavevector. A scalar external potential has the form

$$\delta V(\mathbf{r}) = \delta V_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{r}}, \quad \hat{\rho}_{\mathbf{q}} = \sum_{\mathbf{k}} c_{\mathbf{k}+\mathbf{q}}^\dagger c_{\mathbf{k}}. \quad (20)$$

Translational invariance makes different wavevectors independent, $\chi(\mathbf{q}, \mathbf{q}') = \delta_{\mathbf{q}\mathbf{q}'} \chi(\mathbf{q})$; it does not eliminate finite $\mathbf{q}$. Only $\hat{\rho}_0 = \hat{N}$ is conserved. At finite $\mathbf{q}$ the perturbation is a density wave, and no exact identity says its quasiparticle coupling is unrenormalized.

For elastic scattering on a spherical Fermi surface,

$$q = 2k_F \sin(\theta/2), \quad (21)$$

so physically relevant scattering spans $0 \leq q \leq 2k_F$. A theory that works only at $q = 0$ does not yet predict transport or pairing.

4.2 The quantitative DFPT–many-body matching condition

Using the scalar relations of [sections 2 and 3](#), define the interacting irreducible polarization through a static local kernel,

$$P(q, 0) = \frac{\chi_s(q, 0)}{1 - f_{xc}(q, 0)\chi_s(q, 0)}. \quad (22)$$

Then

$$1 - (v + f_{xc})\chi_s = (1 - vP)(1 - f_{xc}\chi_s). \quad (23)$$

Comparing [equations \(6\) and \(13\)](#) shows that equality between the two scalar vertices requires

$$z\Gamma_\rho(q, 0) \simeq \frac{1}{1 - f_{xc}(q, 0)\chi_s(q, 0)} = \frac{P(q, 0)}{\chi_s(q, 0)}. \quad (24)$$

This is the quantitative relationship among quasiparticle residue, vertex, and static response relevant to the cited electron-gas comparison. It is not the same as setting $z\Gamma_\rho = 1$, and it is not produced by the dynamic Ward identity alone.

Recent vertex-diagrammatic Monte Carlo calculations for the electron liquid at $r_s = 1$ –5 report that the fully interacting quasiparticle vertex and a DFPT-like expression with a static exchange–correlation kernel agree accurately over momenta extending to $2k_F$ [[Cai et al., 2025](#)]. The residual dependence on the incoming fermion momentum is weak, with backscattering providing the most demanding comparison. This is nontrivial numerical evidence for [equation \(24\)](#) across finite momentum; it is not an exact theorem.

4.3 Why the UEG is exceptionally simple

The UEG contains a scalar density channel but no atomic orbitals, sublattices, crystal-field splittings, bond angles, hybridization matrix, Umklapp mixing, or reciprocal-space local-field matrix. Its low-energy perturbation space is therefore effectively one-dimensional. Moreover, local and semilocal density functionals are built from electron-gas information, making the UEG the most favorable setting for a density-kernel closure.

This explains both the value and the limitation of the UEG result:

- it provides a controlled baseline for the total-density channel and a candidate finite- q prior;
- it cannot determine the correction to orbital, bond, hopping, spin, or interaction-modulation channels that do not exist in the model; and
- its finite- q agreement must be tested, not assumed, when transferred to a crystal with local-field and multi-orbital structure.

UEG conclusion. The UEG does not show that real materials have only $q = 0$ coupling. It shows that when the perturbation space contains only density, the many-body problem may be compressed into an accurate static density kernel over a surprisingly wide momentum range. The next-generation theory asks whether an analogous compression exists channel by channel in solids.

5 From a common shift to real material perturbations

5.1 Projection and operator decomposition

Let P_c project onto a localized low-energy or correlated subspace, such as transition-metal d orbitals and ligand p orbitals. The projected DFPT perturbation is

$$D_{\nu q} = P_c \partial_{\nu q} V_{KS} P_c. \quad (25)$$

For each correlated site i , split its on-site block into an orbital trace and a traceless remainder,

$$D_i^{\text{charge}} = \frac{\text{Tr}_{\text{orb}} D_{ii}}{N_{\text{orb}}} I_i, \quad (26)$$

$$D_i^{\text{internal}} = D_{ii} - D_i^{\text{charge}}. \quad (27)$$

The off-site blocks D_{ij} , $i \neq j$, describe changes in hopping and hybridization. If needed, extra channels represent derivatives of interaction parameters, such as $\partial U/\partial u$ and $\partial J/\partial u$. Thus

$$D_{\nu\mathbf{q}} = D_{\nu\mathbf{q}}^{\text{charge}} + D_{\nu\mathbf{q}}^{\text{internal}} + D_{\nu\mathbf{q}}^{\text{nonlocal}} + D_{\nu\mathbf{q}}^{\text{interaction}}. \quad (28)$$

This decomposition is basis-dependent, so the project must state the Wannier or projector construction and test robustness against a reasonable basis change.

5.2 Why a local common operator does not shift all bands equally

Suppose a phonon produces the same local potential D on all orbitals of the correlated subspace. A Bloch state still shifts by

$$\delta\varepsilon_{n\mathbf{k}} = D\langle\psi_{n\mathbf{k}}|P_c|\psi_{n\mathbf{k}}\rangle, \quad (29)$$

which depends on its correlated-orbital weight. At finite momentum,

$$g_{mn}(\mathbf{k}, \mathbf{q}) = D\langle\psi_{m,\mathbf{k}+\mathbf{q}}|P_c|\psi_{n\mathbf{k}}\rangle, \quad (30)$$

which also depends on phases and interband overlaps. DFPT already computes these geometric and hybridization effects. The proposed correction concerns the many-body renormalization of the operator, not a replacement of all matrix elements by a common number.

5.3 Evidence from correlated materials

DFT+DMFT calculations provide a useful controlled contrast because they evaluate the derivative of a local, frequency-dependent correlated self-energy [Abramovitch et al., 2025]. For SrVO_3 with $U = 4.5$ eV and $J = 0.675$ eV, the reported zero-frequency coupling for an M-point Jahn–Teller mode increases from about 44 meV in DFPT to 87 meV in DFT+DMFT. The corresponding R-point breathing-mode coupling changes from about 58 meV to 50 meV. The same correlated material therefore contains a strongly corrected orbital-splitting mode and a comparatively stable charge-like breathing mode.

For hole-doped CaCuO_2 at a representative interaction $U = 3.1$ eV, the reported zero-frequency half- and full-breathing couplings change from approximately 70 to 76 meV and from 53 to 45 meV, respectively [Abramovitch et al., 2025]. These static changes are modest even though the quasiparticle weight is reduced. At larger $U = 4.7$ eV, however, the same study finds strong vertex frequency dependence on the phonon energy scale. Agreement at $\omega = 0$ is then insufficient to validate a dynamic scattering calculation.

These cases support a mode-resolved interpretation, but they do not yet prove the specific decomposition in equation (28). Two further methods establish that standard DFPT has genuine, computable limitations:

- DFPT+ U corrects both the reference electronic structure and the response of Hubbard occupations. In CoO , this changes the polar Fröhlich interaction relative to ordinary DFPT [Zhou et al., 2021].
- GW perturbation theory differentiates a nonlocal, energy-dependent GW self-energy. It produces correlation-enhanced electron–phonon interactions in $\text{Ba}_{1-x}\text{K}_x\text{BiO}_3$ [Li et al., 2019].

These are not competing anecdotes. They identify distinct missing channels: local correlated occupations in DFPT+ U , local dynamic self-energy in DFT+DMFT, and nonlocal quasiparticle self-energy in GW perturbation theory. Earlier correlated-electron analyses likewise show that vertex corrections can be strongly momentum- and channel-dependent [Grilli and Castellani, 1994].

5.4 What “DFPT works in a strongly correlated material” can mean

A material may be strongly correlated according to its spectral weight, Hubbard bands, or mass enhancement while a particular phonon derivative $\partial_u \Sigma^{\text{corr}}$ remains small or nearly common within the relevant low-energy subspace. Conversely, a seemingly small traceless orbital component can be strongly amplified near an orbital instability. The correction is controlled by the product of perturbation weight and channel susceptibility, not by either factor alone.

Accordingly, a successful DFPT result in one mode should not be generalized to all modes of that material. The correct unit of comparison is a material–phonon–momentum–frequency tuple.

6 Working physical hypothesis

The preceding theory suggests a hypothesis that is stronger than a statement of numerical cancellation and weaker than a claimed universal theorem.

Channel-selective protection hypothesis. For a low-energy electron–phonon process, the leading error of static DFPT is controlled by (i) the decomposition of the self-consistent phonon perturbation into charge, internal, nonlocal, and interaction-modulation operators; (ii) the many-body amplification in each operator channel; and (iii) the momentum and frequency path of the response. Strong correlation changes the error mainly when the phonon has appreciable weight in an unprotected channel with a large or dynamic susceptibility.

Write the projected perturbation in an orthonormal operator basis,

$$D_{\nu\mathbf{q}}^{\text{KS}} = \sum_a d_{a,\nu\mathbf{q}} O_a, \quad \text{Tr}(O_a^\dagger O_b) = \delta_{ab}. \quad (31)$$

The corresponding many-body perturbation is represented by a channel kernel

$$D_{\nu\mathbf{q}}^{\text{MB}}(\omega) = \sum_{ab} \mathcal{K}_{ab}(\mathbf{q}, \omega) d_{b,\nu\mathbf{q}} O_a. \quad (32)$$

The first implementation may retain only diagonal or small-block kernels, but the general expression allows channel mixing. The hypothesis predicts the following.

1. The strict uniform common-charge direction has an exact consistency constraint: it is an energy-zero shift and cannot acquire an arbitrary quasiparticle renormalization.
2. The finite- q density channel is not exactly protected, but UEG evidence suggests that it may often be represented by a compact static kernel.
3. Traceless on-site orbital operators, hopping/hybridization operators, and interaction derivatives are not constrained by charge conservation and may receive large, mode-specific corrections.
4. Static and dynamic validity are separate. A channel may have $\mathcal{K}_a(\mathbf{q}, 0) \simeq 1$ while varying rapidly across $\omega_{\nu\mathbf{q}}$.

6.1 Measurable channel weights

A simple Frobenius norm of an operator can overemphasize matrix elements that do not connect the relevant electronic states. For a target energy window $\mathcal{W}$ and nonnegative sampling weights $w_{n\mathbf{k}}$, define a Fermi-window channel strength

$$S_{a,\nu\mathbf{q}} = \sum_{mn\mathbf{k} \in \mathcal{W}} w_{n\mathbf{k}} w_{m,\mathbf{k}+\mathbf{q}} |\langle \psi_{m,\mathbf{k}+\mathbf{q}} | d_{a,\nu\mathbf{q}} O_a | \psi_{n\mathbf{k}} \rangle|^2. \quad (33)$$

Normalize $w_a = S_a / \sum_b S_b$ for interpretation. These weights answer which operators the actual low-energy scattering process samples, not merely which matrix block is largest in a localized representation.

A correction-risk score should combine channel weight and amplification. A minimal form is

$$R_{\nu\mathbf{q}} = \left[\sum_a w_{a,\nu\mathbf{q}} |\mathcal{K}_a(\mathbf{q}, 0) - 1|^2 \right]^{1/2} + \lambda_{\text{dyn}} \max_a \eta_{a,\nu\mathbf{q}}^{\text{dyn}}, \quad (34)$$

where

$$\eta_{a,\nu\mathbf{q}}^{\text{dyn}} = \omega_{\nu\mathbf{q}} |\partial_\omega \ln \mathcal{K}_a(\mathbf{q}, \omega)|_{\omega=0}. \quad (35)$$

The threshold and λ_{dyn} are calibration parameters, not universal constants.

6.2 Falsification criteria

The hypothesis must be rejected or revised if any of the following occur across a held-out benchmark set:

- channel weights do not correlate with the size or sign of the beyond-DFPT correction;
- nominally charge-dominated modes show large, uncontrolled corrections that cannot be explained by finite- q , local-field, or dynamic effects;
- the kernel requires essentially independent parameters for every band-momentum matrix element, eliminating the proposed compression; or
- dynamic effects dominate so broadly that a static risk gate cannot identify a useful low-cost regime.

The theory will then need an enlarged operator basis, nonlocal channel mixing, or a fully frequency-dependent formulation. A negative result would still identify why the proposed simplification fails.

7 A next-generation corrected-DFPT framework

The proposed method is a correction layer, not a replacement for the mature machinery of DFPT. It uses DFPT to obtain phonons, the fully self-consistent Kohn-Sham perturbation, and its Bloch-state geometry, then spends many-body effort only on a low-dimensional projected operator.

7.1 Corrected perturbation and matrix element

In a diagonal-channel approximation, define

$$D_{\nu\mathbf{q}}^{\text{next}}(\omega) = D_{\nu\mathbf{q}}^{\text{KS}} + \sum_a [\mathcal{K}_a(\mathbf{q}, \omega) - 1] D_{\nu\mathbf{q}}^{(a)}. \quad (36)$$

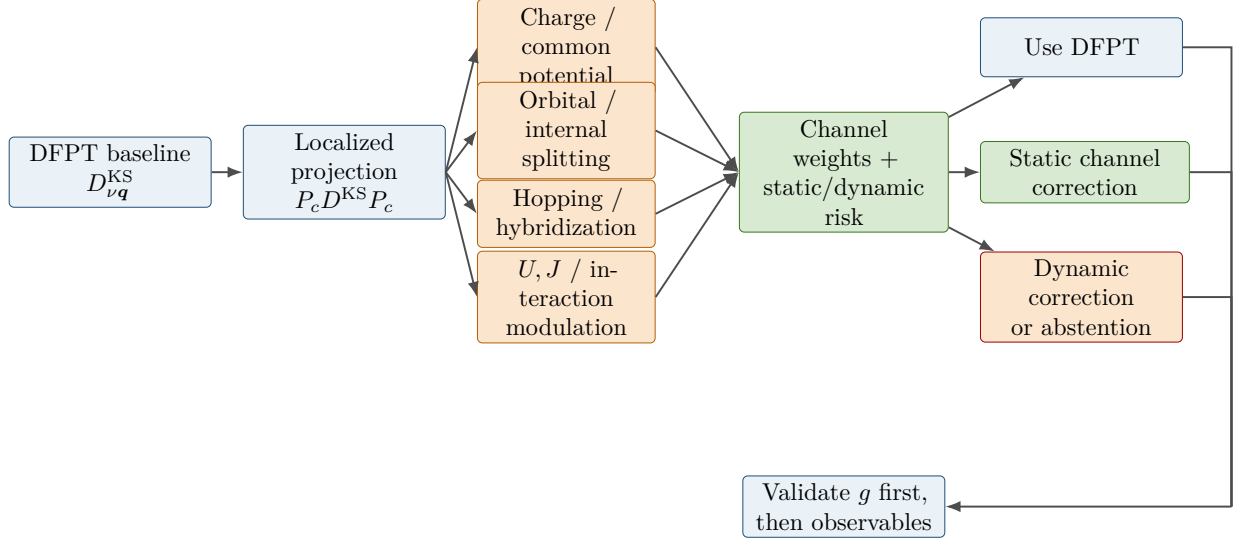


Figure 1: Proposed channel-resolved workflow. DFPT supplies the baseline perturbation. Projection and operator decomposition expose which physical channels a phonon drives. A risk gate selects uncorrected DFPT, a static channel correction, or a dynamic calculation. Validation is performed first on matrix elements and only then on observables.

The predicted quasiparticle coupling is

$$g_{mn\nu}^{\text{next}}(\mathbf{k}, \mathbf{q}, \omega) = \langle \psi_{m, \mathbf{k}+\mathbf{q}} | D_{\nu\mathbf{q}}^{\text{next}}(\omega) | \psi_{n, \mathbf{k}} \rangle. \quad (37)$$

When all $\mathcal{K}_a = 1$, the method reduces exactly to its DFPT baseline. Because the correction acts on operators before Bloch-state matrix elements are taken, it preserves the band, momentum, and phase dependence already present in DFPT.

7.2 How each channel is corrected

Charge channel. The implementation must enforce the exact uniform common-potential constraint and pass a gauge/chemical-potential shift test. At finite q , the UEG relation

$$\mathcal{K}_\rho(q, 0) \approx \frac{P(q, 0)}{\chi_s(q, 0)} \quad (38)$$

is a calibratable prior, not a universal crystal formula. Crystal local-field matrices, dimensionality, polar long-range fields, and multi-band screening must be added explicitly. A practical first model can fit the density kernel to a small number of high-level q points while imposing the exact $q = 0$ anchor.

Local internal channels. For orbital splitting or mixing in a correlated subspace, obtain $\partial_u \Sigma_{ab}(\omega)$ from finite-displacement DFT+DMFT, a linearized impurity response, or a validated low-energy model. If $\eta_a^{\text{dyn}} \ll 1$, retain only $\mathcal{K}_a(\mathbf{q}, 0)$. Near orbital, spin-state, or Mott instabilities, the dynamic kernel and off-diagonal channel mixing must be kept.

Nonlocal channels. When a phonon mainly changes hopping, hybridization, or long-range exchange, a local impurity self-energy is insufficient. Targeted GW perturbation theory, finite-difference quasiparticle calculations, or a nonlocal embedding method can provide selected matrix elements of $\mathcal{K}_{ab}$ [Li et al., 2019]. Symmetry and locality can then interpolate those corrections over equivalent bonds and wavevectors.

Interaction-modulation channels. If a displacement changes U , J , or another downfolded interaction, compute those derivatives with constrained response and include the resulting two-body vertex in the low-energy model. Constrained DFPT provides a precedent for separating screening processes during downfolding [Nomura and Arita, 2015].

7.3 Three-level decision gate

The framework chooses the least expensive level justified by diagnostics.

Table 1: Decision levels for the proposed method.

Level	Physical diagnostic	Calculation	Required check
DFPT-safe	Charge dominated; adequate reference bands; weak estimated dynamics	Standard DFPT matrix elements	Gauge anchor and one representative high-level control
Static correction	Internal or nonlocal weight is appreciable, but η^{dyn} is small	Channel-projected static $\mathcal{K}_a(\mathbf{q}, 0)$	Held-out modes and momenta
Dynamic correction	Rapid vertex variation, soft electronic mode, small coherence scale, or proximity to an instability	Frequency-dependent $\mathcal{K}_{ab}(\mathbf{q}, \omega)$	Values across the physical phonon-frequency window
Abstain	Outside the validated channel/basis/material domain	No automatic prediction	Request a higher-level calculation

7.4 Uncertainty and invariance requirements

Every reported g^{next} should include uncertainty from high-level training data, operator-basis choice, momentum interpolation, and neglected channel mixing. The following tests are mandatory:

- invariance to unitary rotations within a declared degenerate orbital subspace;
- the acoustic sum rule and appropriate translational constraints;
- the uniform scalar-potential gauge test;
- Hermiticity and time-reversal/symmetry relations for D^{next} ; and
- recovery of uncorrected DFPT with $\mathcal{K} = I$.

7.5 Why this is a new prediction framework

Existing beyond-DFPT methods demonstrate how to compute particular corrections, but they remain expensive when applied to all modes and wavevectors. The proposed novelty is the combination of (i) an operator-level physical diagnosis, (ii) a hierarchical choice of correction theory, (iii) a low-dimensional transferable kernel, and (iv) explicit uncertainty and abstention. Its success criterion is not that it matches a full many-body calculation by construction, but that it predicts which parts of the DFPT matrix element need correction and reaches a defined accuracy with far fewer high-level calculations.

8 AI-agent research and implementation program

The AI agent is not used to replace physical validation. Its role is to maintain traceability across literature, material cases, calculations, code, and failed hypotheses, while automating the repetitive parts of comparison and model construction.

8.1 Closed-loop workflow

For each candidate material and phonon mode, the agent executes the following auditable loop.

1. **Read and extract.** Search primary literature and record the material, mode eigenvector, $\mathbf{q}$, electronic method, interaction parameters, definition and normalization of g , frequency at which the vertex is evaluated, and reported uncertainty.
2. **Classify the perturbation.** Project the DFPT perturbation, compute the charge, internal, and nonlocal channel weights of [equation \(33\)](#), and attach a human-readable physical interpretation.
3. **Select a benchmark.** Choose the least expensive many-body method able to represent the active channel: UEG/vertex Monte Carlo for density, DFT+DMFT for local correlated channels, GW perturbation theory for nonlocal quasiparticle response, or an explicit model when interaction derivatives are central.
4. **Generate a hypothesis.** Propose a low-dimensional form of $\mathcal{K}_{ab}(\mathbf{q}, \omega)$ with symmetry, Ward/gauge, locality, and causality constraints encoded before fitting.
5. **Design a discriminating calculation.** Select the mode or momentum point that maximizes disagreement between the current hypothesis and a plausible alternative, instead of merely adding more similar data.
6. **Implement and test.** Produce a versioned correction module, unit tests for algebraic invariants, and numerical regression tests against held-out high-level vertices.
7. **Update or reject.** Store negative evidence and revise the claim ledger. The agent may not promote an empirical relation to an exact principle.

Each extracted number must retain source, equation/table/figure location, normalization convention, and whether it was read directly or inferred. This is essential because electron-phonon matrix elements can differ by phonon normalization, spin factors, mode phase, orbital gauge, and screening convention.

8.2 Minimal benchmark ladder

The proposed ladder is designed to distinguish physical channels, not simply to cover many chemical compositions.

The UEG is the first diagnostic object because it isolates density physics. It must not be the only validation object, because success there cannot test internal crystal channels.

8.3 Accuracy, speed, and transparency metrics

The target should be evaluated against three independent axes.

Table 2: Minimal benchmark ladder for developing and falsifying the method.

System	Active contrast	Reference level	Question answered
Uniform elec- tron gas	$q = 0$ anchor ver- sus $0 < q \leq 2k_F$ density response	Vertex diagrammatic Monte Carlo	Is the finite- q density prior accurate and what controls its residual?
SrVO ₃	Jahn–Teller versus breathing modes	DFT+DMFT	Does traceless orbital weight predict the large mode-selective correction?
Doped CaCuO ₂	Half- versus full- breathing, static versus dynamic vertex	DFT+DMFT	Can the dynamic-risk in- dicator detect failure of a static correction?
CoO	Correlated insula- tor and polar cou- pling	DFPT+ U / experiment	Does the framework detect an invalid reference state and occupation-response channel?
Ba _{1-x} K _x BiO ₃ or a simpler GWPT solid	Nonlocal quasipar- ticle response	GW perturbation theory	Can a sparse nonlocal ker- nel reproduce the high-level correction?

Accuracy. Report the error of complex matrix elements where phases are comparable, squared couplings averaged on the relevant energy shell, and final observables. A practical initial threshold is to reduce the high-level-versus-DFPT error by at least a factor of two while keeping the median corrected $|g|$ error below 10–15% on held-out modes. This threshold is a project criterion and may be revised after the evidence matrix is complete.

Cost. Count the number of high-level displaced/response calculations, total core or GPU hours, memory, and wall time relative to a dense many-body calculation of every mode and $\mathbf{q}$. Speedup is meaningful only at comparable numerical convergence.

Transparency. For every prediction, output the operator decomposition, selected correction level, training-source distance, uncertainty, and any violated invariant. The framework should expose why a mode was corrected rather than only returning a number.

8.4 Implementation layers

The initial software can be separated into small testable modules:

1. a reader for DFPT perturbation matrices and wavefunctions;
2. localized-subspace construction and gauge checks;
3. operator decomposition and Fermi-window channel weights;
4. adapters for high-level reference data from UEG, DFT+DMFT, DFPT+ U , and GWPT;
5. constrained kernel fitting/interpolation with uncertainty;
6. corrected-matrix-element reconstruction; and

7. validation reports separating matrix-element and observable gates.

No production implementation should begin before the two-day literature sprint freezes the first hypothesis, the exact comparison quantities, and the first two material-mode contrasts.

9 Conclusions and immediate decision

The apparent broad usefulness of DFPT should not be described as a mysterious global cancellation. DFPT computes a full, state-dependent static Kohn–Sham response. The missing many-body object is the phonon derivative of the beyond-Kohn–Sham self-energy. That derivative can be small for one perturbation and large for another even when the equilibrium self-energy and mass enhancement are large.

The clearest physical anchor is a uniform common potential coupled to conserved total charge: it only changes the energy zero, so interactions cannot arbitrarily renormalize it. The Ward identity states this constraint precisely, but its dynamic and static limits must not be confused. The UEG extends the picture with strong numerical evidence that a static density kernel remains accurate over finite momenta up to $2k_F$ in the tested regime. Because the UEG contains only the density channel, this result suggests a compression principle rather than a universal proof.

Real crystals add internal orbital, bond, hopping, hybridization, spin, and interaction-modulation operators. Existing DFT+DMFT, DFPT+ U , and GWPT results show that their corrections can be strongly mode-selective. This motivates the proposal’s central hypothesis: the useful small parameter is not “correlation strength” by itself, but the weight of a phonon in each operator channel times the many-body amplification and dynamical variation of that channel.

The proposed next-generation calculation therefore retains DFPT, decomposes its perturbation, and applies a channel kernel only where diagnostics demand it. This can be simpler and faster than computing a complete many-body vertex while being more physically transparent than a black-box correction to every matrix element.

First decision after submission. The first research milestone should test one sharp statement: in SrVO_3 , does a basis-robust measure of traceless orbital-channel weight separate the strongly enhanced Jahn–Teller mode from the weakly corrected breathing mode, while the UEG density relation supplies the charge-channel control? A positive result justifies building the static prototype; a negative result forces the hypothesis to be revised before software expansion.

A Detailed derivations and convention checks

A.1 DFPT response factorization in a scalar system

Starting from [equations \(4\) and \(5\)](#),

$$\delta V_{\text{KS}} = \delta V_{\text{ion}} + (v + f_{\text{xc}})\chi_s \delta V_{\text{KS}}, \quad (39)$$

$$[1 - (v + f_{\text{xc}})\chi_s] \delta V_{\text{KS}} = \delta V_{\text{ion}}. \quad (40)$$

With $P = \chi_s/(1 - f_{xc}\chi_s)$,

$$(1 - vP)(1 - f_{xc}\chi_s) = 1 - f_{xc}\chi_s - v \frac{\chi_s}{1 - f_{xc}\chi_s} (1 - f_{xc}\chi_s) \quad (41)$$

$$= 1 - (v + f_{xc})\chi_s. \quad (42)$$

Thus

$$g^{\text{DFPT}} = \frac{g^0}{(1 - vP)(1 - f_{xc}\chi_s)}. \quad (43)$$

Comparing with $g^{\text{phys}} = g^0 z \Gamma_\rho / (1 - vP)$ yields $z \Gamma_\rho = P / \chi_s$. This algebra establishes the matching condition under the scalar definitions; it does not establish that the condition is exact.

A.2 Uniform-potential cancellation at a quasiparticle pole

Let the pole satisfy

$$E_u - \varepsilon - Du - \Sigma_0(E_u - Du) = 0. \quad (44)$$

Differentiate at $u = 0$:

$$\frac{dE}{du} - D - \partial_\omega \Sigma_0 \left(\frac{dE}{du} - D \right) = 0. \quad (45)$$

Provided the pole is well-defined, $(1 - \partial_\omega \Sigma_0)(dE/du - D) = 0$, so $dE/du = D$. In the alternative form where $\partial_u \Sigma = -D \partial_\omega \Sigma$, the same result is $z(D + \partial_u \Sigma) = D$ with $z = (1 - \partial_\omega \Sigma)^{-1}$. The result follows from the common shift of frequency and chemical potential, not from weak interaction.

A.3 Limit ordering in the Ward identity

Expand the right-hand side of [equation \(14\)](#) for small four-momentum:

$$G^{-1}(k + q) - G^{-1}(k) = i\nu \partial_{i\omega} G^{-1} + \mathbf{q} \cdot \nabla_{\mathbf{k}} G^{-1} + \mathcal{O}(q^2). \quad (46)$$

At $\mathbf{q} = 0$, matching the coefficient of $i\nu$ gives the scalar dynamic vertex. At $\nu = 0$, matching the coefficient of $\mathbf{q}$ gives the longitudinal current vertex. A static scalar perturbation also changes the distribution and self-consistent field, producing the compressibility vertex. This explicit expansion shows why swapping the two limits changes the conclusion.

A.4 Matrix generalization

In a crystal, v , P , χ_s , f_{xc} , and ϵ are matrices in reciprocal vectors $\mathbf{G}, \mathbf{G}'$. Products do not generally commute, so the scalar factorization in [equation \(23\)](#) cannot be transferred by simply replacing every scalar with a matrix. A crystal implementation must retain matrix ordering and local-field components, or derive a controlled projected version of the relation. This is one of the first theoretical tasks of the project.

B Evidence matrix

Table 3: Current evidence for and against the working hypothesis. Numerical values follow the normalization reported in the cited studies and must be checked again before cross-code fitting.

System / mode	DFPT or base-line	Higher-level result	Dominant interpretation	Status and use
Strict uniform common potential	Bare/common shift D	Exact pole shift D	Conserved total charge / energy-zero shift	Exact anchor; no finite- q scattering claim
UEG, $0 \leq q \leq 2k_F$, $r_s = 1-5$	Static DFPT-like density kernel	Vertex diagrammatic Monte Carlo	Single scalar density channel	Recent preprint evidence for finite- q compression; not a theorem [Cai et al., 2025]
SrVO ₃ , M-point Jahn–Teller	about 44 meV	about 87 meV at $\omega = 0$	Traceless orbital splitting	Strong positive example for an unprotected internal channel [Abramovitch et al., 2025]
SrVO ₃ , R-point breathing	about 58 meV	about 50 meV at $\omega = 0$	More charge-like local modulation	Same-material control; modest static correction [Abramovitch et al., 2025]
Doped CaCuO ₂ , half breathing	about 70 meV	about 76 meV at $\omega = 0$	Charge-transfer / bond response	Modest static correction at representative U [Abramovitch et al., 2025]
Doped CaCuO ₂ , full breathing	about 53 meV	about 45 meV at $\omega = 0$	Charge-transfer / bond response	Modest negative static correction at representative U [Abramovitch et al., 2025]
Doped CaCuO ₂ , larger U	Static value may appear moderate	Strong ω dependence on phonon scale	Low coherence scale / dynamic local vertex	Counterexample to using only a static correction [Abramovitch et al., 2025]
CoO, polar coupling	Semilocal DFPT reference is qualitatively deficient	DFPT+ U changes electronic structure and Fröhlich coupling	Correlated occupation response plus polar screening	Established standard-DFPT limitation [Zhou et al., 2021]
Ba _{1-x} K _x BiO ₃	DFT linear response	GW perturbation theory enhances coupling	Nonlocal and energy-dependent quasiparticle response	Established beyond-DFT derivative route [Li et al., 2019]

The matrix is deliberately organized by perturbation type rather than by a single material-wide correlation label. Its central unresolved test is whether the operator decomposition predicts the observed contrast without fitting each mode independently.

C Two-day literature and hypothesis-freeze protocol

The initial exploration is limited to two days. Its purpose is not to survey the entire electron–phonon literature, but to decide whether the channel-selective hypothesis is physically defensible and to freeze the first decisive comparison.

Day 1: establish constraints and collect contrasts

1. **Morning: foundations.** Verify the DFPT response convention, physical quasiparticle vertex, proper versus screened vertex, Ward identity, and static/dynamic limit ordering from primary sources [Baroni et al., 2001, Giustino, 2017, Hedin, 1965, Nambu, 1960].
2. **Midday: UEG.** Reproduce the algebra of equation (24); extract the reported q , r_s , temperature, and error ranges from Cai et al. [2025]; list which statements are numerical and which are exact.
3. **Afternoon: correlated contrasts.** Extract mode definitions, eigenvectors, parameters, normalization, static vertices, and frequency dependence for SrVO_3 and CaCuO_2 .
The primary source is Abramovitch et al. [2025]. Add CoO and one GWPT case to identify failure mechanisms not represented by local DMFT.
4. **Evening gate.** Produce a one-page evidence table with at least one supporting and one challenging case for each proposed channel.

Day 2: turn the idea into a falsifiable first experiment

1. **Morning: define observables.** Freeze the localized subspace, operator normalization, Fermi-window weights, target g convention, and basis robustness checks.
2. **Midday: compare alternatives.** Test at least three hypotheses on the evidence table: correlation strength alone, quasiparticle residue alone, and channel weight times channel susceptibility. Record which observations each cannot explain.
3. **Afternoon: design the first calculation.** Select a SrVO_3 Jahn–Teller/breathing contrast and a UEG density control. Specify input data, code path, acceptance thresholds, and the result that would reject the channel hypothesis.
4. **Final gate.** Freeze one paragraph stating the physical hypothesis, one equation for the first correction model, one benchmark table, and one stop/go decision for implementation.

Required outputs after 48 hours

- a source-checked evidence matrix with no mixed normalization conventions;
- a claim ledger labeling exact results, established methods, numerical evidence, and hypotheses;
- the first two material/mode comparisons and their input availability;
- a minimal operator-decomposition prototype specification; and
- an explicit decision: implement, revise the hypothesis, or stop.

The literature sprint is successful only if it reduces uncertainty about the physical hypothesis. The number of papers read is not itself a success metric.

D Claim ledger: what is known and what is proposed

Table 4: Epistemic status of the central statements.

Status	Statement	Main support
ESTABLISHED	Standard adiabatic DFPT is a self-consistent static Kohn–Sham response and computes the matrix element of $\partial_{\nu q} V_{\text{KS}}$.	Gonze [1995] , Baroni et al. [2001] , Giustino [2017]
ESTABLISHED	The many-body response closes screening, polarization, self-energy, and a vertex; they are not three arbitrary independent factors.	Hedin [1965]
EXACT LIMIT	A strict uniform common scalar potential coupled to total charge is an energy-zero shift; the corresponding dynamic Ward limit gives $z\Gamma_{\omega}^0 = 1$.	Nambu [1960] and appendix A
LIMITATION	The static $\mathbf{q} \rightarrow 0$ density response is not fixed by $z\Gamma = 1$ alone; finite- q density is not a conserved operator.	Ward limit ordering and Fermi-liquid compressibility
NUMERICAL EVIDENCE	In the tested UEG regime, a DFPT-like static density-kernel vertex agrees accurately with a high-level vertex over momenta up to $2k_F$.	Cai et al. [2025] ; recent preprint
NUMERICAL EVIDENCE	Beyond-DFPT corrections can differ strongly between phonons in the same correlated material and can be frequency dependent.	Abramovitch et al. [2025]
WORKING HYPOTHESIS	Channel content times channel-resolved many-body amplification is a better predictor of DFPT error than U or z alone.	Proposed here; to be tested on table 2
PROPOSED METHOD	A small static/dynamic operator kernel can correct DFPT with substantially fewer high-level calculations than a full vertex grid.	section 7 ; not yet validated
OPEN QUESTION	The scalar UEG matching condition has a useful, basis-robust matrix generalization in a crystal with local fields and multiple orbitals.	First formal task after the literature sprint

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